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FLEXURAL FLOW FOLDING AND THE PALEOMAGNETIC FOLD TEST: AN EXAMPLE OF STRAIN REORIENTATION OF REMANENCE IN THE MAUCH CHUNK FORMATION

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Abstract. The relationship between the remanent magnetization and the detailed strain geometry around a first-order fold in the Appalachian Valley and Ridge Province was investigated to examine whether penetrative strains associated with folding can generate a apparent synfolding geometry from a prefolding magnetization. Paleomagnetic results from the Mississippian Mauch Chunk Formation on both limbs of the Frackville Anticline near Lavelle, Pennsylvania, yield two magnetic components, an intermediate unblocking temperature (300°C-600°C) Kiaman remagnetization and a two-polarity high unblocking temperature (650°C-680°C) characteristic magnetization. When the magnetic directions are incrementally corrected for bedding tilt, the intermediate-temperature component is most tightly clustered at 85% unfolding (D = 176°, I = 3°) and the high-temperature component is most tightly clustered at 75% unfolding (D = 184°, I = 27°). Mesoscopic and microscopic structural fabric analyses suggest a strain history that includes a significant component of flexural slip/flow folding. In the coarser-grained sandstone units, folding has largely been accommodated by slip on bedding, while in the finer-grained beds, folding has been accommodated by grain-scale deformation mechanisms such as pressure solution and low-temperature plasticity. Finite strain measurements, determined from center-to-center distances between quartz grains, yield strain ellipsoids consistent with this folding model. Inclination of the characteristic component varies as a function of the magnitude of the finite strain. This variation suggests that the characteristic magnetization has been systematically reoriented with respect to bedding during folding. Remanence directions on the south dipping limb have been rotated to shallower inclinations, while those on the north dipping limb have been rotated to steeper directions causing the prefolding magnetization to appear synfolding. These rotations are in agreement with models of rigid particle rotation in deforming viscous media. Unlike the characteristic magnetization, the secondary component appears to be unaffected by the deformation, and its synfolding behavior is interpreted as the acquisition of a secondary magnetization during Alleghenian folding. These results show that it is important to consider penetrative strains when evaluating the significance of apparent synfolding magnetizations.

INTRODUCTION

Recently, paleomagnetists have used a modification of Graham's [1949] fold test to argue for synfolding magnetizations by noting that the magnetic directions cluster when fold limbs are only partially unfolded [e.g., Scotese and Van der Voo, 1983; Miller and Kent, 1988]. The underlying assumption is that the best-cluster increment of unfolding denotes the position the fold limbs occupied when the rocks acquired their magnetization. However, numerical and geometric modeling of pure shear associated with buckling [Cogne and Perroud, 1985; Cogne et al., 1986] and simple shear associated with flexural slip/flow folding [Facer, 1983; Van der Pluijm, 1986, 1987; Kodama, 1986, 1988] suggests that if the magnetic grains themselves are preferentially reoriented during deformation, a modified prefolding magnetization may mimic a synfolding magnetization. Given these observations, the interpretation that a magnetization was acquired during folding, without consideration for the effects of penetrative strains can lead to erroneous conclusions about both the age and direction of a rock unit's magnetization.

The possibility that deformation could alter the magnetic properties of rocks was first proposed by Graham [1966, p. 627], who showed that the magnetic susceptibility ellipsoids of sedimentary rocks from the Valley and Ridge Province have what he called "systematic relationships with structural setting. These are interpreted as resulting from flow orientation of irregular magnetic grains during a stage of plastic deformation." Sparíosu et al. [1984] showed a decrease in dispersion of site mean directions in the Cherverie Formation in Nova Scotia when these directions were corrected for flexural flow folding. In an extensive study of the Permian-Triassic red beds of the Helvetic nappes, Hirt et al. [1986] found site mean paleomagnetic directions dispersed about a great circle normal to the regional fold axes. They attribute this dispersion to realignment of the magnetic grains by north-south penetrative simple shear associated with nappe emplacement. Kligfield et al. [1981, 1983] and Cogne and Perroud [1985] found that with increasing strain in the Permian red beds of the Alpes Maritimes, hematite grains progressively align themselves parallel to the plane of cleavage. This alignment results in a rotation of the remanent directions.

All of these previous studies are regional in scope and do not specifically test whether rotation of prefolding magnetizations to a synfolding geometry

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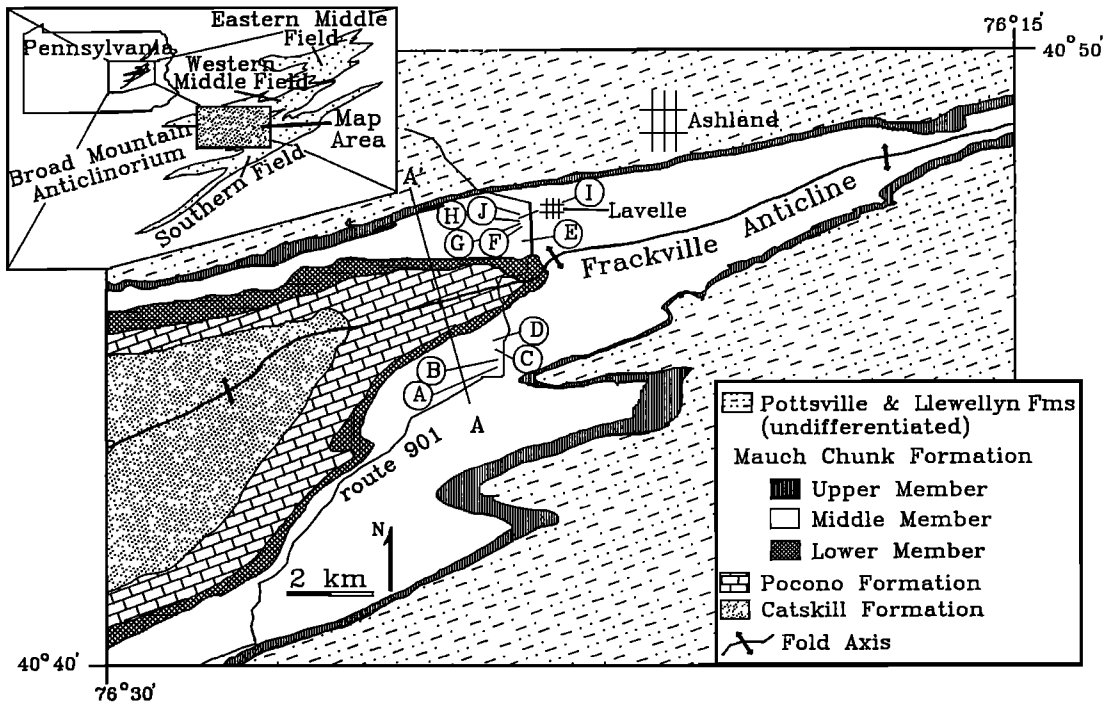


Fig. 1. Generalized geologic map of Frackville Anticline adapted from Arndt [1971a, 1971b], Wood and Trexler [1968], and Wood et al. [1968]. Letters inside circles note the position of sampling sites. Map insert in the top right corner shows the location of the Frackville Anticline relative to the anthracite coal fields in central Pennsylvania. A-A′ designates location of cross section in Figure 6.

as suggested by numerical simulation [Cogne and Perroud, 1985; Cogne et al., 1986; Van der Pluijm, 1986, 1987; Kodama, 1986, 1988] occur in naturally folded rocks. Recent paleomagnetic and structural analysis of the Bloomsburg Formation in the Pennsylvania-Maryland fold and thrust belt suggests that strains associated with folding have modified the prefolding Silurian magnetization to an apparent synfolding, Devonian remagnetization (J. Stamatakos and K. P. Kodama, The effects of grain-scale deformation on the Bloomsburg Formation pole, submitted to Journal of Geophysical Research, 1990). This paper investigates how penetrative strain affects remanence in hematite-bearing sedimentary rocks by examining the relationship between the strain geometry and the magnetizations in the Mississippian Mauch Chunk Formation exposed along both limbs of the Frackville Anticline in the Pennsylvania Valley and Ridge (Figure 1). This fold is particularly well suited for this study because the Mauch Chunk Formation appears to contain a primary or early diagenetic characteristic magnetization [Kent and Opdyke, 1985] and because deformation in the central Appalachian Valley and Ridge includes a significant component of layer-parallel shear associated with flexural slip/flow folding [Faill et al., 1973; Nickelsen and Cotter, 1983; Mitra, 1987].

GEOLOGICAL SETTING AND SAMPLING

The Mauch Chunk Formation consists of gray to red claystone, shale, siltstone and sandstone which accumulated either as flood plain deposits [Wood et al., 1969] or as prograding fluvial-deltaic deposits [Edmunds et al., 1979]. It reaches its maximum thickness of approximately 1500 m near the Broad Mountain region in central Pennsylvania [Wood et al., 1969] (Figure 1) and has been subdivided into three informal members: lower and upper members consisting of interbedded gray and red beds and a middle member consisting almost entirely of grayish red strata [Wood et al., 1969]. Its red color is thought to be primary, the result of oxidation during deposition [Wood et al., 1969]. Near Pottsville, Pennsylvania, the Mauch Chunk-Pottsville contact defines the Mississippian-Pennsylvanian period boundary [Wood et al., 1969]. However, to the north the contact between the two units is transgressive and the Mississippian-Pennsylvanian time boundary lies within the upper Mauch Chunk member [Wood et al., 1969].

The Frackville Anticline is one of the first-order structures that comprise the Broad Mountain Anticlinorium (Figure 1) [Nickelsen, 1963]. It is part of the central Appalachian Valley and Ridge fold and thrust belt formed during the Alleghenian

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Orogeny [e.g., Hatcher et al., 1989]. At the outcrops exposed along route 901 near Lavelle, Pennsylvania, the middle member of the Mauch Chunk is exposed on both limbs, but the hinge has been eroded to expose the underlying Pocono Sandstone. Paleomagnetic samples were collected from 10 sites, four on the south dipping limb and six on the north dipping limb. Sites on each limb were located 5-10 m apart. At each site, located within a single bed, eight samples, covering a stratigraphic thickness of approximately 1 m, were collected. Sites were chosen from beds with various grain sizes in order to test for magnetization differences in rocks with different structural competence. All paleomagnetic samples were drilled in the field using a portable gasoline-powered drill fitted with a 2.5 cm diameter diamond coring bit. Each sample was oriented using a standard paleomagnetic orientation devices and a magnetic compass. Cleavage and bedding orientations were made in the field with a Brunton compass with an estimated accuracy of 3°.

METHODS

Progressive thermal demagnetization (12-15 steps) was performed on all 80 samples collected from this fold using a Schonstedt TSD-1 thermal demagnetizer. Remanence measurements were made on a two-axis CTF cryogenic magnetometer at Lehigh University. Characteristic magnetizations were determined from principal component analysis [Kirschvink, 1980]. The distribution of sample directions around each of the site mean directions was calculated using Fisher statistics [Fisher, 1953]. The hypothesis that the directions from both limbs originated from a single population was tested by comparing the precision parameters on each limb [McFadden and Lowes, 1981]. Incremental fold tests were performed using 5% increments of limb dip unfolded about the line of strike. The statistical significance of the fold test was determined following the methods suggested by McFadden and Jones [1981], using the site mean directions.

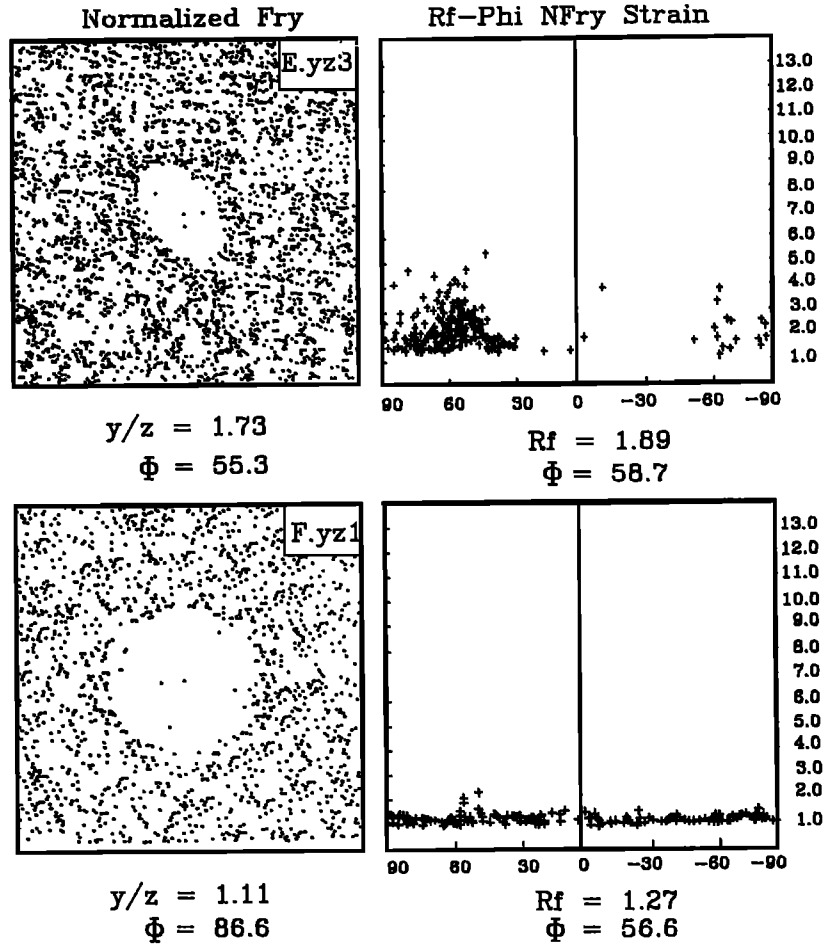


Fig. 2. Representative normalized Fry (NFry) and Rf-ϕ diagrams showing variation between sites with low and high strains. Each diagram is derived from the center-to-center distance of 120-150 quartz grains. Both strain ellipses are the YZ principal plane of the strain ellipsoid. ϕ is the orientation of the long axis of the strain ellipse, in degrees relative to horizontal.

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Finite strain was measured using center-to-center and object strain techniques [Erslev, 1988]. Oriented thin sections were then photographed and projected onto a digitizing tablet where between 120 and 250 quartz grains per sample were digitized. Two least squares, best fit ellipses were calculated for each sample, one from the enhanced, normalized Fry diagram [Fry, 1979; Erslev, 1988] and one from an Rf-Ø plot [Dunnet, 1969] based on the normalized center-to-center data (Figure 2). The normalization procedure reduces variations due to object size and sorting by dividing the center-to-center distance between two objects by the sum of their average radii [Erslev, 1988]. Enhancement eliminates subjectivity of manually fitted Fry [1979] ellipses by calculating a least squares best fit ellipse from only those points within the rim of maximum point density [Erslev, 1989]. In the Mauch Chunk samples, quartz grain pairs with a center-to-center distance less than 1.1 times the sum of their radii were used to calculate the best fit ellipses. Rf-Ø object ellipses for each grain were generated by best-fit ellipses through the centers of the 10 nearest neighboring grains.

Finite strains were determined for six of the 10 sites, three on each limb of the fold. At each site, three independent measurements of the finite strain were averaged to get a site mean finite strain. In order to determine the finite strain ellipsoids, two orthogonal thin sections were cut parallel to the principal planes. The first set was cut parallel to cleavage, which was assumed to be the XY principal plane [Wood, 1974]. From this first set of measurements, the orientations of the other two principal planes were determined, and a second set of thin sections was cut parallel to the YZ principal plane. In order to estimate the strain heterogeneity, measurements were repeated in three different regions of the same thin section and compared to the between-site variations.

To test the assumption that the initial quartz grain distributions were random, the Rf-Ø values for one sample from each site were unstrained using Peach and Lisle's [1979] algorithm based on Lisle's [1977] "theta-curve" method. This procedure incrementally "unstrains" the final Rf-Ø distribution by superimposing a reciprocal coaxial strain ellipse. At each stage, the randomness of the 'undeformed' quartz grain center distribution is checked by a χ^2 test.

RESULTS

Paleomagnetic results. Similar to previous paleomagnetic results from the Mauch Chunk [Knowles and Opdyke, 1968; Kent and Opdyke, 1985; Kent, 1988], thermal demagnetization uncovered two magnetic components: a B component with distributed unblocking temperatures between 300°C and 660°C and a characteristic C component with discrete unblocking temperatures between 665°C and 680°C (Figure 3). The B component was isolated in 76 of 80 samples, and B component site mean directions were calculated for all 10 sites (Table 1). The C component was isolated in 56 of the 80 samples, which enables C component site

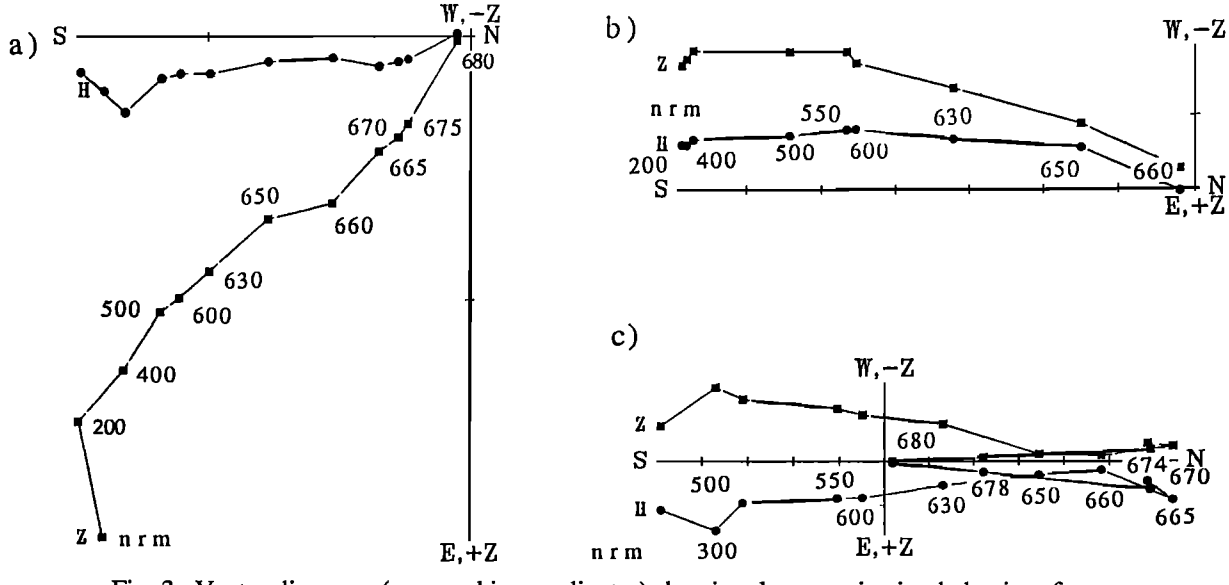


Fig. 3. Vector diagrams (geographic coordinates) showing demagnetization behavior of the Mauch Chunk samples; (a) sample from the south dipping limb; (b and c) samples from the north-dipping limb. For Figures 3a and 3c, two components were isolated, the B component (200°C to 600°C) and the C component (665°C to 680°C). For Figure 3b, only the B component was isolated (400°C to 660°C). H, horizontal component; Z, vertical component. Ticks on axes are in units of milliamperes per meter. Numbers on curve give temperature in degrees Celsuis.

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TABLE 1. Paleomagnetic Results From the Frackville Anticline

	C Component							B Component						
Site	N/No	Ig	Dg	Is	Ds	k	α95	N/No	Ig	Dg	Is	Ds	k	α95
A	8/8	59	197	26	179	18.9	11	8/8	34	184	-1	179	48.3	7
B	5/8	44	177	8	172	54.4	9	8/8	32	171	-5	169	38.9	8
C	6/8	52	212	24	192	29.7	11	7/7	32	186	-3	189	47.6	8
D	8/8	50	186	15	176	26.0	10	7/8	32	177	-4	174	109.1	5
E	8/8	17	189	39	187	18.5	12	7/8	-15	172	6	173	65.0	7
F	7/8	-1	12	-23	12	51.2	7	8/8	-10	173	10	174	46.8	7
G	0/8	-	-	-	-	-	-	8/8	-14	179	7	178	58.0	7
H	0/8	-	-	-	-	-	-	8/8	-14	180	7	180	68.5	6
I	0/8	-	-	-	-	-	-	8/8	-12	178	9	178	79.9	6
J	7/8	-10	8	-32	7	23.6	11	7/8	-22	169	-1	171	49.0	8
Limb means														
South 4/4	52	197	18	178	49.0	10	4/4	33	179	-4	176	181.2	5	
North 3/6	-10	10	-31	9	94.8	8	6/6	-15	175	6	176	204.1	4	

N/No is number of samples used in analysis/number of samples collected and demagnetized; Ig(s) is the inclination in geographic (stratigraphic) coordinates; Dg(s) is the declination in geographic (stratigraphic) coordinates; k is the Fisher[1953] precision parameter; and α95 is the Fisher [1953] 95% confidence radius. Site coordinates are 40.7°N, 283.6°E.

mean directions to be calculated for seven of the 10 sites (Table 1).

The B component, on the south dipping limb of the fold, has a south and moderately downward direction in geographic coordinates which becomes south and flat in stratigraphic coordinates. On the north dipping limb of the fold the B component has a south and shallowly upward direction in geographic coordinates which becomes south and shallowly downward in stratigraphic coordinates (Table 1, Figure 4a). The two limb mean directions have statistically similar precision parameters [McFadden and Lowes, 1981], indicating that the B component site mean directions originated from a single population of directions. Incremental unfolding of the B component around strike yields directions that are most tightly clustered at 85% unfolding (Figure 5). This distribution is significantly different at the 95% confidence level from both the prefolding and postfolding configurations [McFadden and Jones, 1981].

The C component directions for all four sites on the south dipping limb of the anticline are south and moderately to steeply downward in geographic coordinates and south and shallowly downward in stratigraphic coordinates (Table 1, Figure 4b). However, site mean directions from the coarser sandy beds appear to be steeper than those from the finer-grained shale beds. On the north dipping limb of the fold, two of the sites have mean directions that are north and shallowly upward in geographic coordinates and north and moderately upward in stratigraphic coordinates (Table 1, Figure 4b). The other northern limb sites have a south and shallowly downward direction in geographic coordinates and a south and moderately downward direction in stratigraphic coordinates (Table 1, Figure 4b). On this limb of the fold, site mean directions from the coarser sandy beds appear to have shallower inclinations that the finer-grained shale beds. The precision parameters for each limb-mean direction are statistically similar [McFadden and Lowes, 1981] indicating that the site mean directions originally

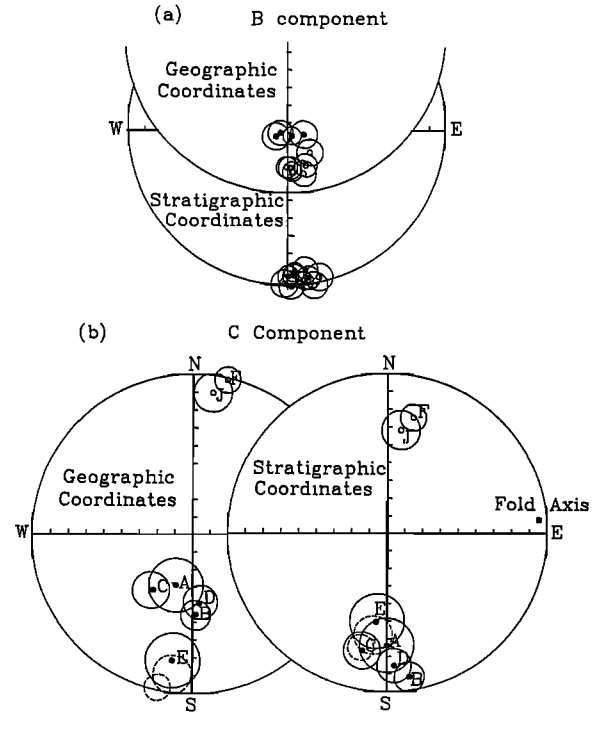


Fig. 4. Equal-area projections showing site mean directions for (a) the B component in geographic and stratigraphic coordinates and (b) the C component in geographic and stratigraphic coordinates. Open (solid) symbols plotted on upper (lower) hemisphere. Dashed circles are the projections of sites F and J in the lower hemisphere.

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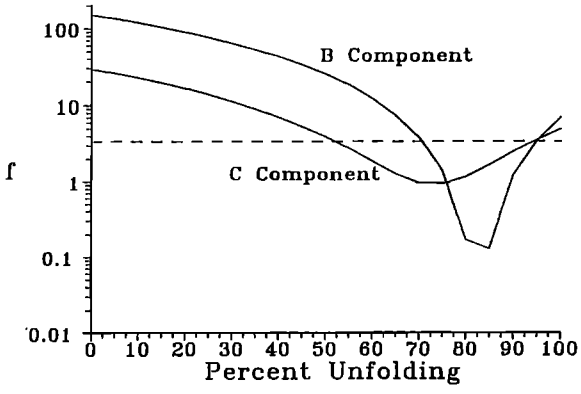


Fig. 5. Variation in McFadden and Jones [1981] parameter f as a function of incremental unfolding of the two fold limbs. Dashed line represents the critical value of f at the 95% confidence level.

belonged to a single population. Incremental unfolding of the C component site means, assuming that the two north and upward directions are opposite polarity directions, yields directions that are most tightly clustered at 75% unfolding (Figure 5), which is significantly different from both the prefolding and post folding configurations at the 95% confidence level [McFadden and Jones, 1981].

Finite strain results. Normalized Fry diagrams (Figure 2) show moderately to well defined circular to elliptical vacancy fields. Rf-Ø plots (Figure 2) show symmetrical distributions of points about the maximum f. For most of the samples with strain ratios greater than 1.50 the Rf-Ø fields appear to be closed. The example shown in Figure 2a yields a fluctuation of approximately 58° [Ramsay and Huber, 1983, p. 77]. For samples with finite strain ratios less than 1.50, the Rf-Ø fields appear to be open, with 180° fluctuations as shown in Figure 2b. Comparison of the Rf-Ø and normalized Fry methods indicates that the two techniques are in good agreement, although the Rf-Ø results tend to yield slightly higher strain ratios and shallower ellipse axes relative to bedding than the normalized Fry results (Table 2). In addition, both the between-sample and within-sample strain variations are small. Standard deviations of the between-sample and within-sample strains, for both the magnitude and orientation of the strain ellipse, average around 10% of the mean value (Table 2).

All samples "unstrained" by the Peach and Lisle [1979] algorithm exhibited random pretectonic orientations at the 90% significance level. The reciprocal strain ratios which produced the lowest χ^2 values were slightly smaller than the ratios obtained from the Erslev [1988] Rf-Ø technique.

Finite strain results from the thin sections cut parallel to cleavage yield X axes of the strain ellipsoids nearly parallel to the strike of cleavage. The Y axes of the strain ellipsoids for all sites except site J are oriented parallel to the trace of cleavage on the YZ principal plane (Figure 6). Site J's strain ellipse, in the YZ plane, is oriented subparallel to bedding.

Comparison of the B component inclination, in stratigraphic coordinates, against the magnitude of the finite strain (Figure 7a) shows that this secondary component does not vary as a function of finite strain. Similarly, no trend is observed when the difference between the site mean inclinations and site A's inclination (Δinc), relative to the sense of shear, is plotted against finite strain (Figure 7b). The change in inclination is plotted in this manner because in flexural slip/flow folds, the sense of shear is opposite on the two fold limbs and, as evident from the finite strain results, site A is relatively undeformed. Using the data in Figure 7b, a best fit linear regression of Y/Z on Δinc yields a correlation coefficient of 0.19, which is not significant at the 95% confidence level.

TABLE 2. Finite Strain Results from the Frackville Anticline

			NFry				Rf-ϕ				
Site	Pl	N	Ra	± σ	ϕ(°)	± σ(°)	Rh	Ra	± σ	ϕ(°)	± σ(°)
A	XY	3	1.10	0.05	1.8	23.8	1.20	1.21	0.03	1.9	19.3
B	XY	3	1.36	0.18	4.9	34.6	1.37	1.39	0.21	7.1	8.0
D	XY	3	1.19	0.16	-29.0	8.4	1.35	1.38	0.05	-15.8	4.4
E	XY	3	1.30	0.15	-5.6	14.1	1.38	1.40	0.02	10.1	8.5
F	XY	3	1.15	0.26	-3.1	53.1	1.12	1.14	0.05	7.8	10.6
J	XY	3	1.49	0.22	-21.0	22.0	1.44	1.49	0.12	4.6	12.1
A	YZ	3	1.24	0.19	61.6	14.3	1.19	1.22	0.06	42.5	20.2
B	YZ	3	1.73	0.06	56.5	4.2	1.70	1.76	0.06	50.0	17.3
D	YZ	3	1.52	0.18	54.0	13.2	1.36	1.38	0.05	54.4	17.6
E	YZ	3	1.68	0.11	-57.4	28.0	1.57	1.63	0.21	-64.7	10.4
F	YZ	3	1.15	0.07	-88.1	3.5	1.25	1.27	0.06	-85.0	9.9
J	YZ	3	1.47	0.07	-29.8	11.0	1.43	1.46	0.14	-13.1	7.4

NFry is normalized Fry; Pl is the principal plane; Ra is the arithmetic mean of axial ratio of strain ellipse; Rh is the harmonic mean of axial ratio of strain ellipse; ϕ is the orientation of long axis of strain ellipse (positive is counterclockwise from horizontal toward the west); and σ is 1 standard deviation around the mean values.

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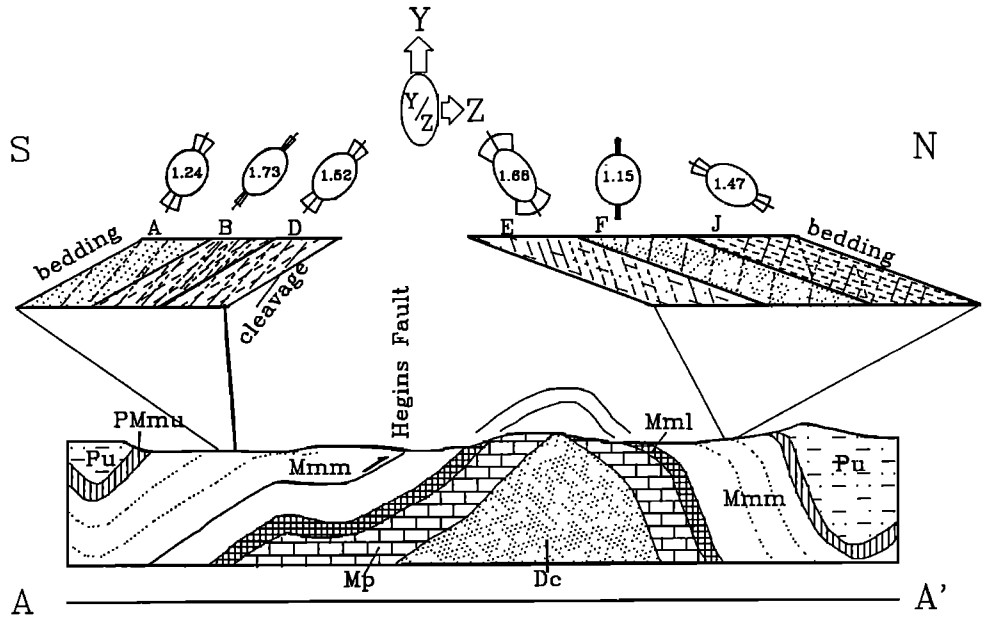


Fig. 6. Cross section of the Frackville Anticline and diagrammatic profile of the two fold limbs showing the orientation and ellipticity of the YZ principal strain ellipse for each site. Arcs around the long axes of the strain ellipses are +/- one standard deviation of the axes' mean orientation. Numbers inside ellipses are the mean axial ratio for each site. Dc, Catskill Formation; Mp, Pocono Formation; Mml, lower middle member Mauch Chunk Formation; Mmm, middle member Mauch Chunk Formation; PMmu, upper member Mauch Chunk Formation; Pu, Llewellyn and Pottsville formations undifferentiated.

In contrast, the C component inclinations vary systematically as a function of finite strain (Figure 7c). The low strain sites (A and F) yield inclinations between 24° and 29°. The more strained sites on the south dipping limb of the fold (sites B and D) have inclinations up to 15° shallower than the low-strain sites, while those on the north dipping limb (sites J and E) have inclinations up to 15° steeper than the low-strain sites. Using the data in Figure 7d, a best-fit linear regression of Y/Z on Δinc yields a correlation coefficient of 0.95, which is significant at the 95% confidence level.

Structural analysis results. The Frackville Anticline (Figure 1) is an open, upright fold with a kink band [Faill, 1973] profile (Figure 6) and a nearly vertical axial surface. Kamb [1959] contouring of 229 poles to bedding taken from the Tremont [Wood and Trexler, 1968], Minersville [Wood et al., 1968], Ashland [Arndt, 1971a], and Mount Carmel [Arndt, 1971b] 7.5' quadrangles (Figure 8) delineates an interlimb angel of 75° and a fold axis that trends N85°E with a 5° plunge (Figure 8).

At the outcrop on the fold's south dipping limb the shale and claystone beds contain a strongly developed spaced cleavage that strikes roughly east-west and dips moderately to steeply to the south. In the coarser-grained sands and siltstones there is moderately developed spaced cleavage with slightly steeper orientations. Correcting the poles to cleavage for bedding tilt significantly reduces their dispersion, suggesting that cleavage formation initiated prior to folding. Petrographic examination indicates that the mesoscopic cleavage is a disjunctive pressure-solution feature in the coarser-grained sandy units, but in the claystones and shales it is a well-developed crenulation cleavage that modifies an earlier fabric oriented at a low angle to bedding. Many of the crenulations are asymmetric and occasionally cut by microfaults. Both microfolds and microfaults suggest top to the north shear. Bedding-parallel faults are common and are marked by quartz-chlorite shear fibers that also indicate top to the north shear. In addition, many large detrital phyllosilicate grains appear to be squeezed and bent around the quartz clasts. Several mudstone beds also contained deformed mud cracks and deformed reduction spots elongated parallel to strike.

At the outcrop on the north dipping limb, cleavage is not as pervasive as it is at the south dipping limb outcrop, and several different orientations were noted. In most of the fine-grained siltstone and shale beds, cleavage is nearly perpendicular to bedding and often forms pencils. In beds bounded by bedding-plane faults this cleavage is more pronounced and dips toward the north relative to bedding. In a few of the finest grained beds, cleavage dips shallowly to the south relative to bedding which may be an artifact of an earlier, prefolding phase of deformation (M. B. Gray, personal communication, 1990). Petrographic examinations of the layer-normal and north dipping cleavage cleavages show that they are

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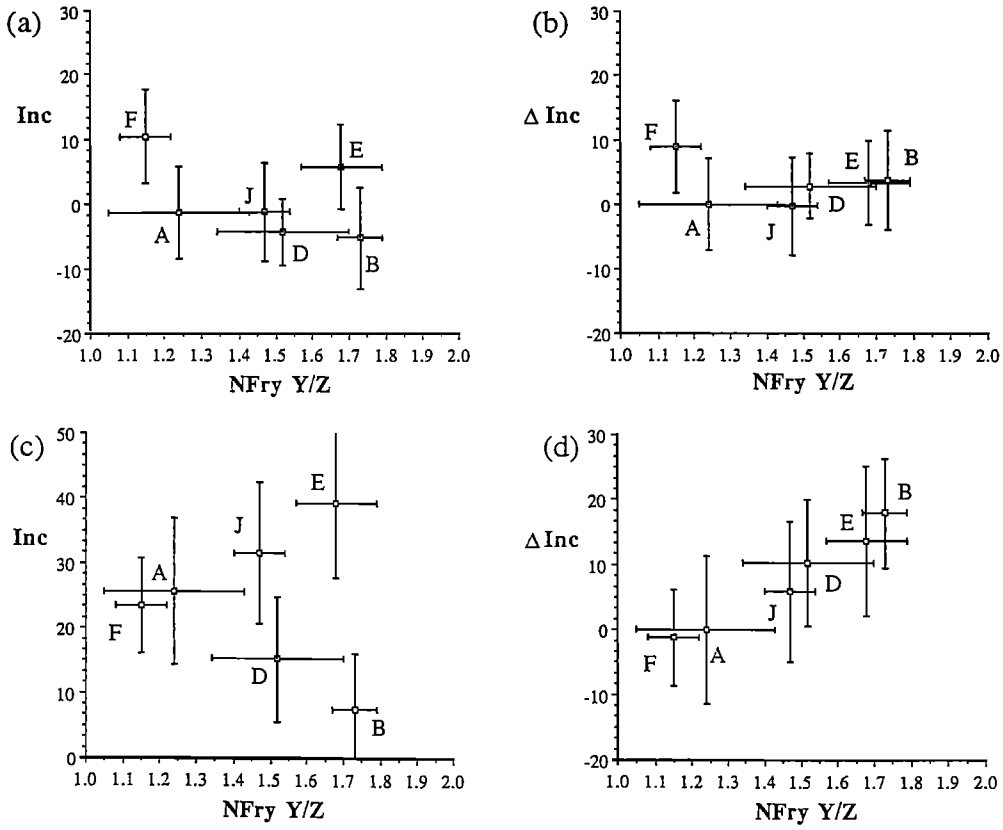


Fig. 7. Plot of the Y/Z axial ratio of the strain ellipsoid against (a and c) site mean inclinations (B and C components) and (b and d) difference between site mean inclinations and site A's inclination (Δinc), relative to the sense of shear (B and C components). Error bars denote +/- one standard deviation in the strain measurement and the Fisher [1953] 95% confidence radii around the inclination.

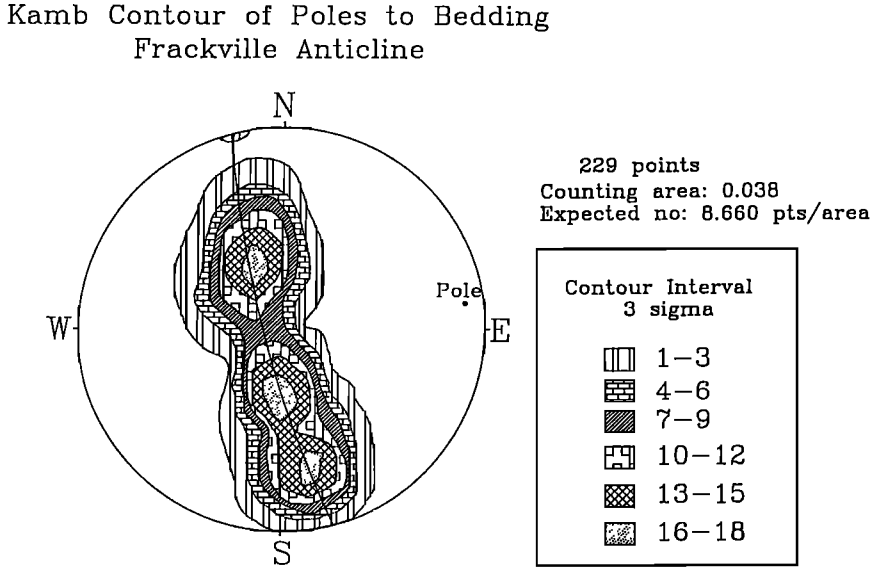


Fig. 8. Kamb [1959] contour of 229 poles to bedding taken from the Tremont [Wood and Trexler, 1968], Minersville [Wood et al., 1968], Ashland [Arndt, 1971a], and Mount Carmel [Arndt, 1971b] quadrangle maps. The pole to the best fit great circle defines the fold axis trending N85°E with a 5° plunge to the east.

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crenulation fabrics accompanied by pressure solution. Quartz-chlorite shear fibers on the bedding plane faults indicate top to the south shear, which is opposite that observed on the south dipping limb.

DISCUSSION

Analysis of finite strain. The two main assumptions in using center-to-center methods as a measurement of finite strain are that the original particle distribution was uniform and that the ellipsoidal particles did not have an initial preferred orientation [Ramsay, 1967]. Systematic redistribution of an initially nonuniform distribution of grain centers will not produce elliptical vacancy fields [Ramsey, 1967]. Because the Mauch Chunk samples yield fairly well defined circular to elliptical vacancy fields, their initial distribution must have been fairly uniform. Analysis of the Rf-Ø diagrams shows that for the higher-strain samples, the distribution of points is not significantly asymmetric and for the low-strain sites, the orientation of elliptical particles is random (Figure 2). In addition, the X^2 results also indicate that the initial distributions of quartz grain centers were random [Peach and Lisle, 1979]. These results suggest that there was no initial preferred orientation of elliptical particles prior to deformation.

It is also important to assess how well these measurements estimate the tectonic strains accumulated in the rocks. The observed mesoscopic and microscopic fabrics suggest that the dominant deformation mechanisms are pressure solution and low-temperature crystal plasticity associated with the development of the crenulated phyllosilicates [e.g. Gray and Durney, 1979; Platt and Vissers, 1980]. With pressure solution and microfolding, the distance between grain centers is modified in proportion to the longitudinal strains along the directions of those strains [Ramsay and Huber, 1983]. Pressure selectively dissolves and reprecipitates quartz in the rocks which allows the more insoluble platy minerals to rotate parallel to cleavage [e.g., Kerrich and Allison, 1978; Engelder and Marshak, 1985]. In the Mauch Chunk rocks, finite strain measurements yield strain ellipses oriented nearly parallel to the trace of cleavage on the outcrop face with larger axial ratios in beds with more intensely developed cleavage (Figure 6). In addition, variations in the magnitude and orientation of the strain ellipse, both within an individual sample and within a site are small. This small variation indicates that at the paleomagnetic-sample scale the finite strain is relatively homogeneous. These results suggest that the strains measured by the center-to-center techniques are representative of the tectonic strains accrued in these rocks.

The orientation of the finite strain ellipses is consistent with bedding-parallel shear associated with flexural slip/flow folding, top-to-the-south shear on the north dipping limb and top-to-the-north shear on the south dipping limb (Figure 6). These shear strains are pervasive on the south dipping limb and more locally developed on the north dipping limb. The only bed where cleavage and the finite strain ellipse do not coincide is bed J on the north limb. The finite strain ellipse from site J is orientated between cleavage and bedding which suggests that the assumption that cleavage is the XY principle plane may not be entirely valid for this site. Based on limb dip [Ramsay, 1967, page 393], folding should yield maximum axial ratios of 2.6 for the south dipping limb and 2.0 for the north dipping limb. Axial ratios for the YZ plane (Table 2, Figure 6), which is the principal plane perpendicular to the flexural slip/flow, are smaller. These smaller than predicted axial ratios suggests that the total flexural strain may have been partitioned between grain-scale processes and slip on bedding and that the measured finite strains are composites of flexural flow and earlier deformations.

Figure 9 plots the three-dimensional finite strain data on a Flinn diagram [Flinn, 1958] along with a hypothetical strain path for the progressive deformation similar to that proposed for the lithologically similar Bloomsburg Formation in Central Pennsylvania [Nickelsen and Cotter, 1983] and the Permian red beds in the Alpes-Maritimes [Graham, 1978; Kligfield et al., 1981]. In this path: (1) Initial compaction generated oblate ellipsoids parallel to bedding. (2) Layer-parallel shortening (LPS) moved the ellipsoids first toward the field of apparent constriction and then back toward the field of apparent flattening. During the second phase of LPS the X axis of the strain ellipsoid was alogned parallel the fold axis (the tectonic Y direction), and Y axis of the strain ellipsoid was aligned normal to bedding (the tectonic X direction) [Ramsay and Huber, 1983]. In this case, LPS stopped short of producing uniaxial oblate ellipsoids. (3) Flexural slip/flow, especially if accompanied by volume loss associated with pressure solution, continued to move the ellipsoid into the field of apparent flattening. Because the X axis of the strain ellipsoid is parallel to strike, the plane perpendicular to the the fold axis is the YZ principal plane. This also explains why the deformed mud cracks and reduction spots appear to be extended parallel to strike.

Analysis of paleomagnetic directions. At the Frackville Anticline the B component appears synfolding at 85% unfolding and yields a mean direction of D = 176°, I = 2° (a_{95} = 3°, k = 214.9), which corresponds to a north paleopole at 47°N, 109°E (dm = 3°, dp = 2°). The prefolding direction for the B component at the Frackville Anticline yields a similar direction of D = 176°, I = 2° (a_{95} = 5°, k = 92.6) which corresponds to a north paleopole at 48°N, 110°E (dm = 5°, dp = 2°). Based on the 30 Ma window reference apparent polar wander path for North America [Irving and Irving, 1982], these paleomagnetic poles fall between the 260 and 240 Ma mean paleopoles for North America. The distributed unblocking temperatures up to 660°C indicate that the B component resides in the small pigmentary hematite, 0.1-1.0 μm in diameter [Dodson and McClelland-Brown, 1980; Walker et al., 1981].

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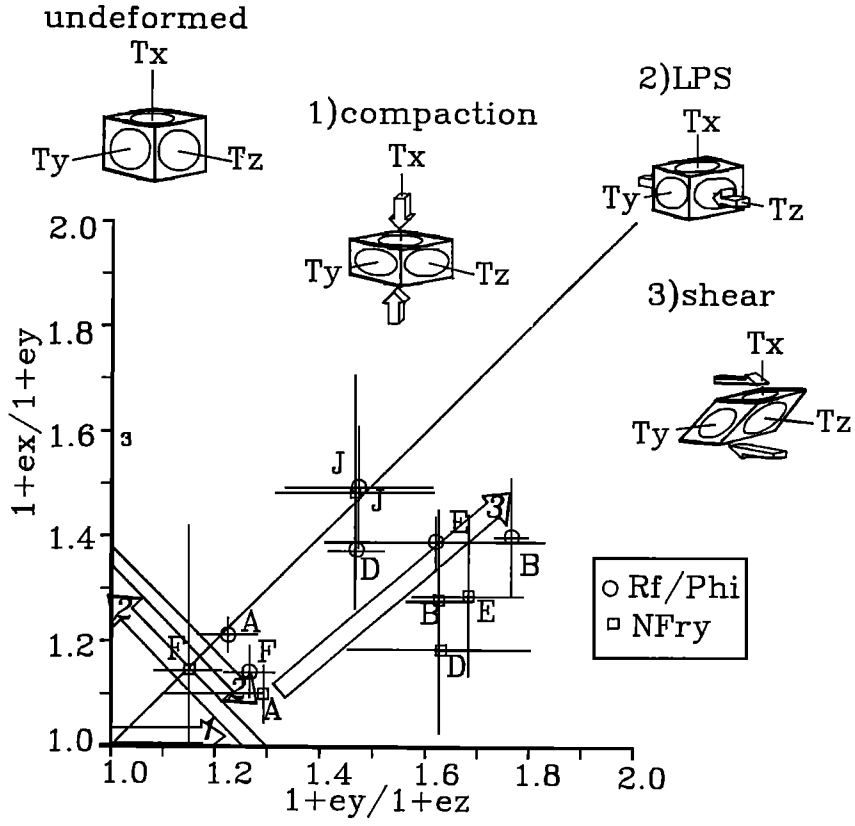


Fig. 9. Flinn [1958] graph of the three-dimensional strain data showing a strain path that moves from low-strain triaxial ellipsoids into the field of apparent flattening. Also shown is a hypothetical strain path (arrows). Boxes show the strain ellipsoids generated during the progressive deformation. Tx, Ty, and Tz are the tectonic axes.

These results coupled with its uniform reversed polarity suggest that the B component is a secondary magnetization acquired during the late Paleozoic Kiaman reversed superchron, possibly as the result of orogenic fluid expulsion during Alleghenian deformation [McCabe and Elmore, 1989; Kent, 1988; Miller and Kent, 1988; Oliver, 1986; Kent and Opdyke, 1985]. The lack of a correlation between the inclination and finite strain (Figures 7a and 7b), and the synfolding nature of much of the Kiaman remagnetizations throughout the Appalachians [McCabe and Elmore, 1989] suggest that the B component's synfolding geometry resulted from the acquisition of a secondary magnetization during Alleghenian folding.

The high unblocking temperatures, above 650°C, indicate that the C component resides in larger, 1-50 μm hematite particles [Dodson and McClelland-Brown, 1980]. The C component appears synfolding and yields a mean direction at 75% unfolding of D = 184°, I = 27° (a95 = 7°, k = 68.2). This mean direction corresponds to a north paleopole at 35°N, 97°E (dm = 7°, dp = 4°). This direction is significantly different from both the mean direction reported by Kent and Opdyke [1985] for all Mauch Chunk sites, D = 160°, I = 28° (a95 = 8°), and the mean direction for Mauch Chunk sites north of the Pennsylvania salient, D = 166°, I = 26° (a95 = 10°) [Kent, 1988, Kent and Opdyke, 1985]. In the Kent and Opdyke [1985] study, which included Mauch Chunk sites distributed all over Pennsylvania, the C component contained numerous reversals suggesting that it is an early diagenetic or detrital magnetization.

The rocks exposed at the Frackville Anticline show penetrative deformations that suggest that the Mauch Chunk's C component has been reoriented by strain to appear synfolding. Like the results from the Permian-Triassic red beds of the Helvetic nappes, [Hirt et al., 1986], the site mean directions, in stratigraphic coordinates, are dispersed along a great circle normal to the fold axis (Figure 4b). Furthermore, there is a good correlation between inclination and the amount and orientation of finite strain (Figures 7c and 7d). On the south dipping limb of the anticline, the south and downward directions are rotated to shallower directions. On the north dipping limb of the anticline, the north and upward and south and downward directions are rotated to steeper directions. Because the remanent vectors on each limb have been rotated in a sense opposite to the rigid rotation of the beds during

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folding, full correction for bedding tilt overcorrects the directions on both limbs, and the prefolding magnetization now appears synfolding.

In the Kent and Opdyke [1985] study, the B component was most tightly grouped at 50% unfolding and the C component was most tightly grouped at 100% unfolding. There are several possible explanations for the differences between the results from Kent and Opdyke [1985] and those presented in this study. Results presented here are from 10 beds located in two limbs of one fold, which in Kent and Opdyke's [1985] sampling scheme would constitute only 2 of 23 sites. The 85% and 75% unfolding values from this study are specific to the Frackville Anticline. The 50% and 100% unfolding values from the previous work are averages for many folded and unfolded sites throughout the Pennsylvania Valley and Ridge. In the Kent and Opdyke [1985] study, finer-grained beds with penetrative cleavage were probably avoided during sampling. In this study samples were collected over a range of grain sizes, including the more deformed finer-grained beds.

If sample directions from the two sites with lowest strain, A and F, are incrementally unfolded, they are most tightly grouped 90% unfolding (Figure 10), but this improvement in grouping over the prefolding configuration is not significant [McFadden and Jones, 1981]. The mean direction from these two sites, D = 186°, I = 25° (a95 = 10°, k = 12.1), overlaps with the mean Mauch Chunk direction for sites north of the Pennsylvania salient [Kent, 1988; Kent and Opdyke, 1985].

The correlation between the inclination of the C component and strain ellipticity and the poor correlation between the inclination of the B component and strain ellipticity suggests that the grains carrying the C component of remanence experienced rotation while the particles carrying the B component remained fixed relative to bedding during folding. The main difference between the two magnetic components is the size of the remanence-carrying grains. The larger hematite grains, which are dispersed in the rock's matrix, act like the rigid porphyroblasts and rotate systematically in response to the deformation. In contrast, the smaller, pigmentary hematite grains appear to be attached to the surface of the quartz grains. Because quartz is deforming via pressure solution, the quartz grains do not rotate relative to bedding and the small hematite flakes appear unaffected by the deformation.

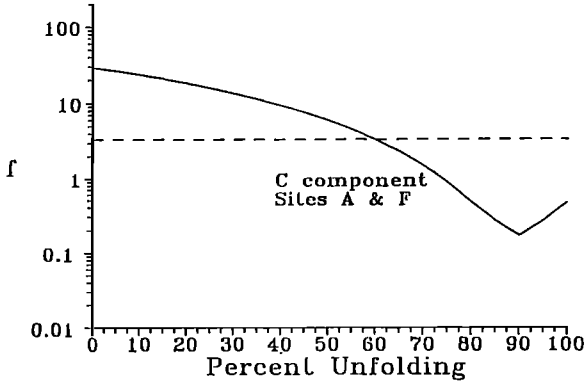


Fig. 10. Incremental unfolding of the C component of the two sites with lowest finite strain (A and F). Dashed line represents the critical value of f at the 95% confidence level. Note that although f reaches a minimum at 90% unfolding, the distribution is not significantly different from the prefolding configuration.

The sense of rotation of the C component vectors also suggests that during deformation the hematite flakes behave as rigid particles in a viscously shearing matrix [Jeffrey, 1923; Gay, 1969] rather than as passive markers [March, 1932]. Hematite is hexagonal with rhombohedral symmetry [Collinson, 1983]. Its remanence lies within the crystal's basal plane [Nagata, 1961], so that any rotation of the basal plane will result in a similar rotation of its magnetization. For passive line behavior the basal plane will be pulled toward the XY principal plane of the strain ellipsoid and once aligned with this principal plane, will simply rotate with it as deformation proceeds. For rigid particle behavior the basal plane will rotate in the same sense as the shear. Given a mean inclination of -25° from the low-strain sites, A and F, all the site mean directions from the more highly strained sites have been rotated in the same sense as the bedding-parallel shear and either through or away from the long axis of the strain ellipse (Figure 11).

The inherent assumption in this hematite rotation model is that the remanence vector remains fixed relative to the basal plane during deformation. The magnetism in hematite results from a slight canting of coupled, antiparallel moments in the A and B sublattices [Dzyaloshinski, 1958]. Lattice realignment during deformation may alter the orientation of this spin canted moment and lead to the rotation of the C component remanent magnetization observed in the Mauch Chunk rocks. However, high-pressure X ray diffraction experiments [Yagi and Akimoto, 1982] show that there are no major changes in the hematite crystal structure below 900 kbar except for a 6-10% volume reduction between 500 and 600 kbar. Given these results, lattice reorientation seems unreasonable, but future work is needed to directly test this assumption.

CONCLUSIONS

Structural analysis in the Mauch Chunk Formation exposed on both limbs of the Frackville Anticline indicates a progressive deformation history that includes a significant component of bedding-parallel shear associated with folding. Finite strain measurements yield strain ellipsoids consistent with flexural flow and suggest that this stage in the deformation sequence has rotated the preexisting fabric elements, including the characteristic

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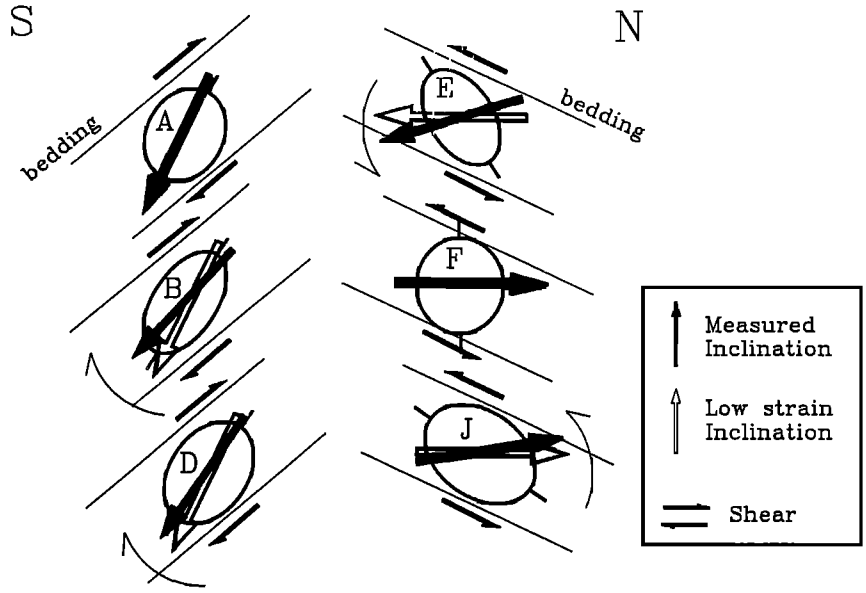


Fig. 11. Profile of the fold showing the orientation of the finite strain ellipses in the YZ principal plane and the orientation of the measured inclination and the mean inclination of sites A and F. Note that the measured inclinations appear to have rotated in the same sense as the shear and either away from or past the long axes of the strain ellipses, which is indicative of rigid particle not passive marker behavior.

magnetization, in a sense opposite to the rigid rotation of the fold limbs. These rotations result in shallower remanent inclinations on the south dipping limb, steeper remanent inclinations on the north dipping limb, and a apparent synfolding geometry from a prefolding magnetization. The sense of this rotation also suggests that the larger hematite grains, which carry the C component, behave as rigid particles not passive markers during deformation. Unlike the characteristic magnetization, the B component remanence, which is carried by the smaller pigmentary hematite flakes, appears to be unaffected by the deformation. Its synfolding geometry is probably the result of a secondary magnetization acquired during Alleghenian folding.

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